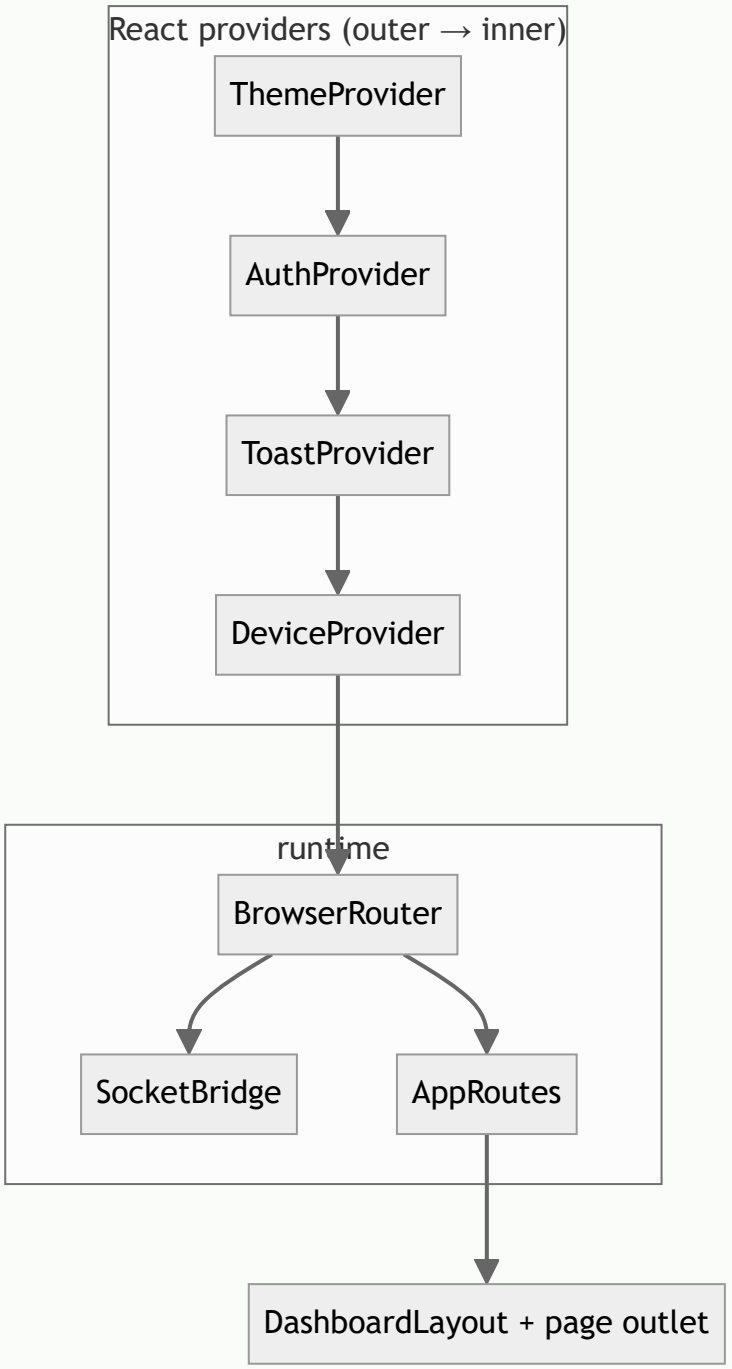
Smart AgriTech EMS Documentation

Web frontend reference

React dashboard for Smart AgriTech EMS: `web_frontend/`.

Built with **Vite**, **React 18**, **React Router**, **Tailwind CSS**, and **Recharts**. Served in production as static files behind **nginx** (Docker image).

Application shell



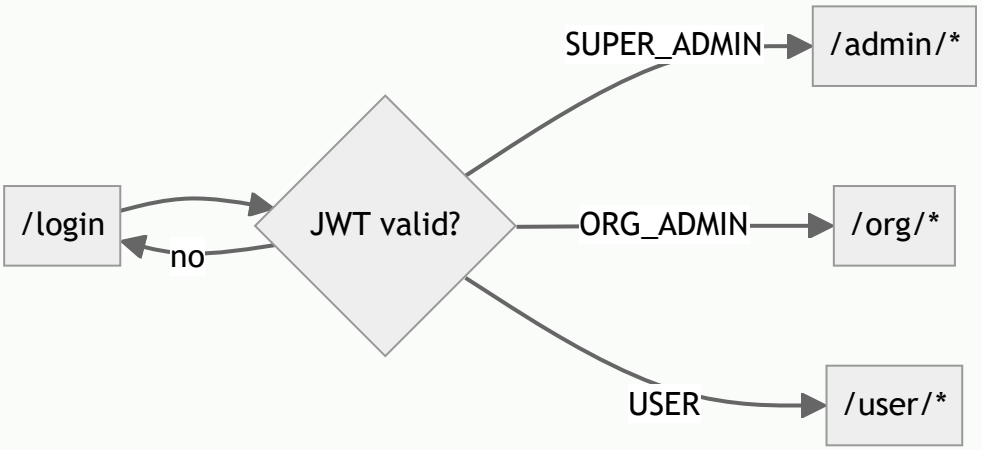
Provider	Responsibility
`ThemeContext`	Light/dark mode, org theme colors from API
`AuthContext`	JWT session, login/logout, role mapping
`ToastContext`	Global success/error toasts
`DeviceContext`	Selected device/slave, device list cache

Provider	Responsibility
`SocketBridge`	Connects Socket.IO when logged in; refreshes devices on `reading:new`

Role-based routing

Backend roles map to URL prefixes via `utils/roles.js`:

Backend enum	Frontend path	Nav config
`SUPER_ADMIN`	`/admin`	`adminNav`
`ORG_ADMIN`	`/org`	`orgNav`
`USER`	`/user`	`userNav`



`ProtectedRoute` redirects unauthenticated users to `/login` and wrong-role users to their home prefix.

Complete route map

Public

Path	Page	Purpose
`/login`	`Login.jsx`	Email/password login
`/`	redirect	→ role home or `/login`

Super Admin — `/admin`

Path	Component	Functionality
`/admin`	`AdminDashboard`	Fleet KPIs, charts, device summary
`/admin/dashboard-detail`	`DashboardDetailPage`	Live variable tiles for selected device
`/admin/organizations`	`AdminOrganizations`	CRUD organizations
`/admin/users`	`AdminUsers`	CRUD all users
`/admin/gateways`	`AdminGateways`	Gateway CRUD
`/admin/devices`	`AdminDevices`	Device list, create, assign template
`/admin/devices/:deviceId`	`DeviceDetailPage`	Slaves, variables, users, MQTT modal
`/admin/device-templates`	`AdminDeviceTemplates`	Template library
`/admin/device-templates/:templateId`	`TemplateDetailPage`	Slaves + variables editor
`/admin/icons`	`AdminManageIcons`	Icon asset management
`/admin/products`	`AdminProducts`	Product catalog
`/admin/data-center`	`AdminDataCenter`	Bulk data operations
`/admin/sensor-history`	`SensorHistoryPage`	Raw ingest log, pagination
`/admin/historical-data`	`AdminHistoricalData`	Multi-variable charts + CSV
`/admin/variable-alarms`	`AdminVariableAlarms`	Active variable alarm states
`/admin/alarm-history`	`AlarmHistoryPage`	Historical alarms (device filter)
`/admin/linkage-records`	`AdminLinkageRecords`	Alarm linkage audit
`/admin/template-triggers`	`AdminTemplateTriggers`	Alarm trigger templates
`/admin/alarm-settings`	`AdminAlarmSettings`	Per-device alarm thresholds
`/admin/alarm-contacts`	`AdminAlarmContacts`	Notification contacts
`/admin/device-timestamps`	`AdminDeviceTimestamps`	Connectivity events
`/admin/schedule-tasks`	`AdminScheduleTasks`	Cron-style device switches
`/admin/theme-settings`	`AdminThemeSettings`	Global UI themes
`/admin/settings`	`AdminSettings`	System settings

Organization Admin — `/org`

Path	Component	Functionality
`/org`	`OrgDashboard`	Org-scoped dashboard
`/org/dashboard-detail`	`DashboardDetailPage`	Live device detail
`/org/users`	`OrgUsers`	Team users in org
`/org/widget-templates`	`OrgWidgetTemplates`	Dashboard widget layouts
`/org/devices`	`OrgDevices`	Org device management
`/org/devices/:deviceId`	`DeviceDetailPage`	Device config (org scope)
`/org/gateways`	`OrgGateways`	Org gateways
`/org/device-templates`	`OrgDeviceTemplates`	Org templates
`/org/device-templates/:templateId`	`TemplateDetailPage`	Template editor
`/org/sensor-history`	`SensorHistoryPage`	Sensor logs
`/org/historical-data`	`OrgHistoricalData`	Historical charts
`/org/device-timestamps`	`OrgDeviceTimestamps`	Connectivity log
`/org/template-triggers`	`OrgTemplateTriggers`	Alarm triggers
`/org/alarm-settings`	`OrgAlarmSettings`	Alarm config
`/org/alarm-contacts`	`OrgAlarmContacts`	Contacts
`/org/alarm-history`	`AlarmHistoryPage`	Alarm log
`/org/schedule-tasks`	`OrgScheduleTasks`	Schedules
`/org/settings`	`OrgSettings`	Org preferences

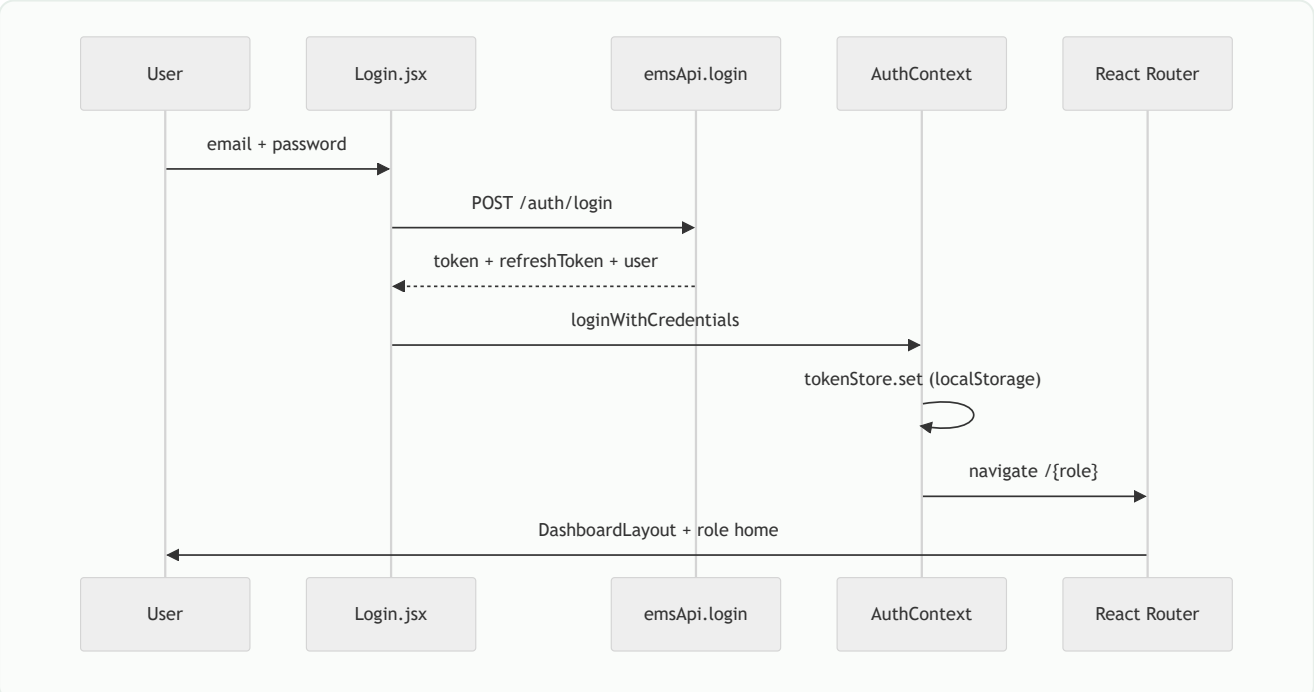
End User — `/user`

Path	Component	Functionality
`/user`	`UserDashboard`	Assigned devices overview
`/user/dashboard-detail`	`DashboardDetailPage`	Live readings
`/user/account`	`UserAccountSettings`	Profile, password
`/user/notifications`	`UserNotifications`	In-app notifications
`/user/subscription`	`UserSubscription`	Plan request
`/user/products`	`UserProducts`	Browse products

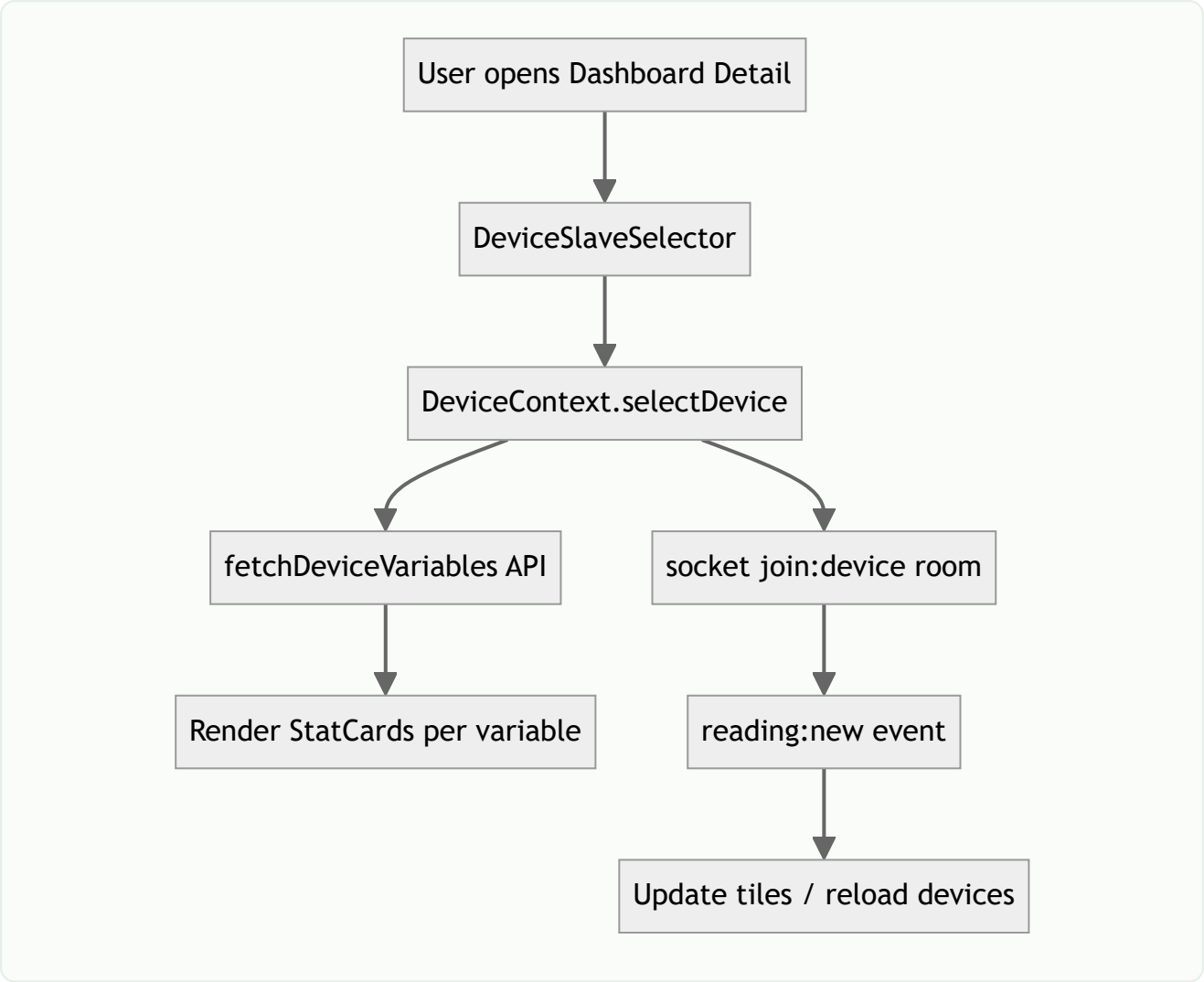
Path	Component	Functionality
`/user/schedule`	`UserSchedule`	View schedules
`/user/slab-rates`	`UserSlabRates`	Electricity tariff slabs
`/user/interval-history`	`UserIntervalHistory`	Billing intervals
`/user/sensor-history`	`SensorHistoryPage`	Sensor history
`/user/alarm-template`	`UserAlarmTemplate`	User alarm templates
`/user/alarm-settings`	`UserAlarmSettings`	User alarm config
`/user/alarm-history`	`AlarmHistoryPage`	Alarm history
`/user/ai-analytics`	`UserAIAnalytics`	AI analytics hub
`/user/voltage-imbalance`	`UserVoltageImbalance`	Phase voltage analytics
`/user/current-imbalance`	`UserCurrentImbalance`	Current imbalance
`/user/power-factor`	`UserPowerFactor`	Power factor trends
`/user/energy-consumption`	`UserEnergyConsumption`	Consumption charts
`/user/anomalies`	`UserAnomalies`	Anomaly timeline

User journey diagrams

Login → dashboard



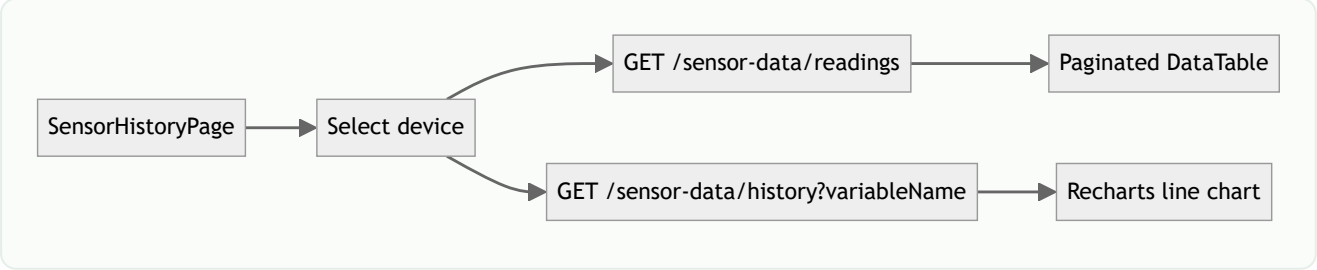
Device selection → live readings



Shared pages (`DashboardDetailPage` , `SensorHistoryPage` , `AlarmHistoryPage`) use:

- `DeviceSlaveSelector` — pick device (and optional Modbus slave)
- `utils/sensorReadings.js` — `fetchDeviceVariables` , `fetchDeviceDashboardCharts` , `latestToReadings`
- `utils/dashboardHelpers.js` — chart aggregation helpers

Sensor history flow



Note: History queries omit `slaveId` when raw ingest rows have null `deviceConfigSlaveId` (common for single-slave devices).

Alarm history flow

API client layer

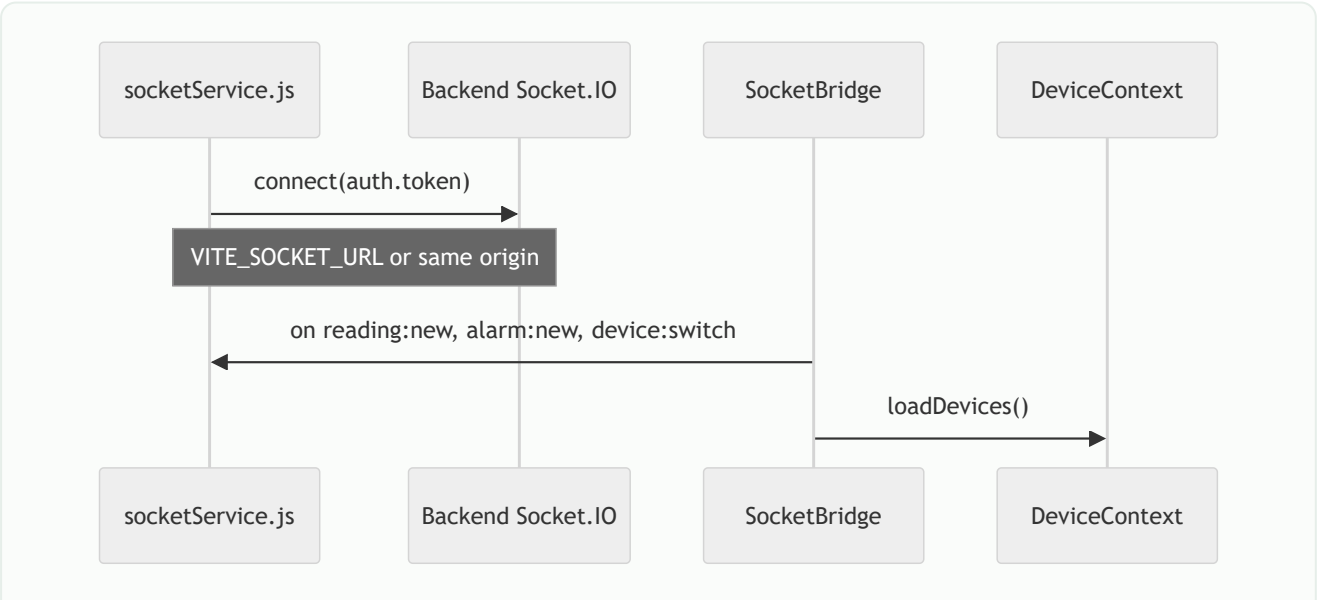
`api/client.js`

Feature	Behavior
Base URL	`VITE_API_URL` (build-time) or `/api` (dev proxy)
Auth header	`Authorization: Bearer <token>`
401 handling	Auto refresh via `/auth/refresh`; logout on failure
Errors	`ApiError` with HTTP status

`api/emsApi.js`

Central facade for all REST calls grouped by domain (auth, devices, sensor-data, alarms, AI, etc.). Pages import `emsApi` rather than raw `fetch`.

Real-time (Socket.IO)



Env var	Production value
`VITE_API_URL`	`https://iotbackend.domain.com/api`
`VITE_SOCKET_URL`	`https://iotbackend.domain.com`

CapRover: enable **WebSocket Support** on the backend app.

Layout & UI components

```
web_frontend/src/
├── components/
│   ├── layout/       DashboardLayout, Sidebar, Topbar
│   ├── shared/       DeviceSlaveSelector, TimeRangeChips
│   ├── ui/           Modal, DataTable, StatCard, FormFields, PageState
│   └── SocketBridge.jsx
├── pages/
│   ├── admin/        Super-admin screens
│   ├── org/          Org-admin screens
│   ├── user/         End-user screens
│   └── shared/       Cross-role pages
├── context/          Auth, Theme, Device, Toast
├── config/navConfig.jsx
└── utils/            roles, mappers, sensorReadings, dashboardHelpers
```

`DashboardLayout` renders sidebar from `navConfig`, top bar (user menu, theme toggle), and `<Outlet />` for nested routes.

Build & environment

Local development

```
cd web_frontend
npm install
npm run dev    # http://localhost:5173
```

Vite dev server proxies `/api` to `http://localhost:5000` (see `vite.config.js`).

Production build

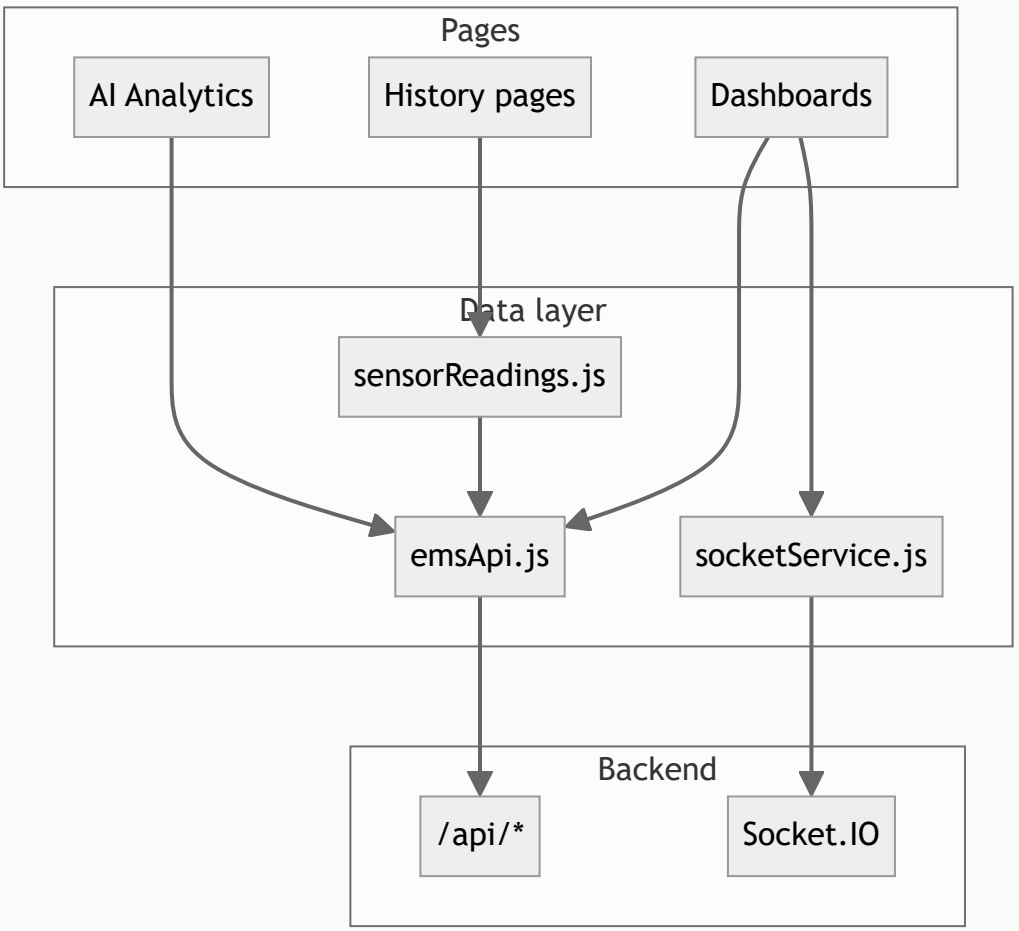
```
VITE_API_URL=https://iotbackend.yourdomain.com/api \
VITE_SOCKET_URL=https://iotbackend.yourdomain.com \
npm run build
```

Output: `dist/` → copied into nginx Docker image.

Docker (`web_frontend/Dockerfile`)

1. Node stage: `npm ci` + `npm run build` with build args
2. nginx stage: serve `dist/` on port 80
3. `nginx.conf` : SPA fallback (`try_files`), gzip, cache static assets

Data flow summary



Role capability matrix (web)

Feature area	Admin	Org	User
Organizations	✓	—	—
All users	✓	org only	self
Gateways / devices CRUD	✓	✓	view assigned
Device templates	✓	✓	—
Icons / products / themes	✓	—	products view
Data center / linkage	✓	—	—
Sensor & historical data	✓	✓	✓
Alarms (full config)	✓	✓	limited
Schedule tasks	✓	✓	view
AI analytics	—	—	✓
Billing (slabs / intervals)	—	—	✓
Notifications	—	—	✓

Related documents

- Application functionality
- Backend API
- Deployment
- Architecture